Industrial Product Intelligence

Turning scraps of supplier data into product information a business can actually sell from

What this document is	A plain-English explanation of the project: the problem, what was built, how it works, what has been proven, and what remains.
Who it is for	Anyone. No technical background is assumed. Every specialist term is explained where it first appears.
Status	A working prototype, live on the internet and continuously monitored.
Live link	industrial-product-intelligence.onrender.com
Submission deadline	23 August 2026

1. The problem, in everyday terms

Imagine a shop that sells 200,000 industrial parts — bearings, valves, motors, sensors. Nobody buys these on impulse. An engineer needs to know that a part is *exactly* the right size, rated for the right pressure, made of the right metal. If the listing is wrong, they order the wrong part, and a machine somewhere stops working.

Now look at where that shop's information actually comes from. One supplier sends a neat spreadsheet. Another sends a PDF catalogue. A third has a website and nothing else. A fourth sends an email that says, in full:

> **A real example of what arrives**
>
> “6205-2RS, SKF, deep groove ball bearing”

That is three pieces of information. A usable product listing needs around twenty: the internal diameter, the external diameter, the width, the load it can carry, the top speed it can spin at, the material, the seal type, and so on. Somebody has to find all of that and type it in.

Today that somebody is a person. They open the manufacturer's catalogue, look up the part, and copy the numbers across. It takes roughly ten minutes per product. For 200,000 products that is about sixteen years of one person's working life.

And people make mistakes. They read a figure from the wrong row. They type millimetres where the catalogue meant inches. They swap two numbers around. Most of those mistakes are invisible — the listing looks perfectly normal, and the error is only discovered when the wrong part arrives on a factory floor.

What this project does

It takes whatever scraps of information a supplier provides and turns them into a complete, checked product listing — automatically. Just as importantly, it **checks its own work**, and it **shows where every single number came from**.

2. The idea at the heart of it

Most systems that fill in missing information are built to always produce an answer. Ask them anything and they will confidently tell you something. That is fine when the stakes are low — a wrong film recommendation costs nothing.

Here the stakes are not low. So this system is built on the opposite instinct:

> **The rule everything else follows from**
>
> A wrong specification on an industrial part stops a machine. So the system would rather leave a box empty, raise a warning, or refuse to publish a product altogether than state something it cannot back up.

In practice that means three habits, which run through everything:

- **Every number carries a receipt.** Nothing appears without a note saying where it came from — which supplier field, which page of which PDF, or which published engineering standard.
- **Confidence is stated, not implied.** A figure taken straight from an international standard is marked as near-certain. An educated guess is labelled a guess, and is visibly flagged for someone to confirm.
- **Bad data is stopped, not smoothed over.** When the numbers contradict each other, the product is blocked from publication and the contradiction is explained in plain language.

Why that matters commercially

A system that is right 95% of the time but does not tell you which 5% is wrong is close to useless, because a human still has to re-check everything. A system that is right 61% of the time and is *honest about the other 39%* is genuinely useful: staff can trust the confident parts and spend their time only where they are needed.

3. What it actually looks like

The prototype is a website. It has four sections, each answering a different real-world situation.

Section 1 — A single product

You type in whatever you have and press a button. Using the three-word example from earlier, the system returns:

What it worked out	Value	Where it came from
Internal diameter	25 mm	International standard ISO 15
External diameter	52 mm	International standard ISO 15
Width	15 mm	International standard ISO 15
Load rating	14 kN	International standard ISO 15
Maximum speed	16,000 rpm	International standard ISO 15
Seal type	Rubber sealed	Decoded from the part number
Ring material	Chrome steel	Typical value — flagged to confirm

Eleven pieces of information from three. And critically, the last row is *marked differently* from the others, because it is an assumption rather than a fact.

> **The trick behind those first five rows**
>
> The code “6205” is not a random part number. It is a designation defined by an international standard, a bit like a postcode. Any bearing labelled 6205, from any manufacturer in the world, has a 25 mm bore and a 52 mm outer diameter. The system knows how to read that code — so those numbers are not guesses at all, they are lookups.

Section 2 — A technical document

Suppliers often provide a datasheet: a PDF with a table of specifications. You drop the file in and the system reads it.

This is harder than it sounds, because datasheets are laid out in wildly different ways. Some have proper tables with lines and borders. Some just line the numbers up with spaces. Some are written as “Bore diameter .... 25 mm” running down the page. The system handles all three, and every value it reads stays linked back to the document it came from, so it can be checked later.

Section 3 — A whole catalogue

Upload a spreadsheet of hundreds of products and every row is processed at once. You get a summary: how many are ready to publish, how many need a human to look at them, how many are blocked, and why. This is the realistic way the tool would actually be used.

Section 4 — Learning

This is the most unusual part, and it gets its own section later.

4. How it works, step by step

When a product goes in, it passes through eight stages. Think of it as a production line where each station does one job and hands the work to the next.

Stage	What happens, in plain terms
1. Tidy up	Clean up the messy input. Recognise that “skf”, “SKF Group” and “S.K.F.” are the same manufacturer. Find the part number even if it is buried in a sentence.
2. Identify	Work out what kind of product this is — a bearing? a valve? a motor? This matters because it decides which questions are worth asking. You do not ask a bearing about its voltage.
3. Read	Pull out every fact already present in the input, converting units as it goes. If the supplier wrote “5 HP” and the catalogue standard is kilowatts, it converts — and records that it did.
4. Fill gaps	Work out what is still missing. Some gaps are filled from engineering standards; some require judgement, which is where the AI comes in.
5. Resolve conflicts	When two sources disagree, prefer the better-evidenced one, record that there was a disagreement, and *lower* the confidence — because a disagreement is itself a reason to be less sure.
6. Write the listing	Produce the customer-facing title, description and bullet points, using only facts that survived the earlier stages.
7. Check	Run the checks. Is anything impossible? Anything contradictory? Anything unsafe?
8. Score	Give the record a mark out of 100 and a verdict: publish, review, or blocked.

The checking stage deserves a closer look

Most quality checks look at one field at a time: is this number in a sensible range? Those catch typos, and little else. The interesting errors are the ones where every individual number looks completely reasonable, and only the *combination* is impossible.

Three real examples the system catches:

What arrived	Why it is impossible
A valve made of PVC plastic, rated to operate at 180°C	PVC softens at about 60°C. Both figures look normal on their own; together they describe a product that would melt.
A bearing with a 90 mm hole and a 40 mm outside diameter	The hole is bigger than the object. Two numbers were swapped during data entry — a completely ordinary mistake that no single-field check would ever notice.
A hydraulic hose rated to burst at only twice its working pressure	The international safety standard requires a four-times margin. This is not untidy data; it is a hose that should not be sold for that job.

There are sixteen such checks, each written from real engineering rules. They are the difference between a system that tidies data and one that actually protects the buyer.

5. The system teaches itself new categories

Here is the obvious objection to everything above. The system knows about ten kinds of product. A real catalogue has hundreds. So what happens when something arrives that nobody prepared it for?

The wrong answer is to hire people to write out hundreds more categories by hand. The answer built here is different: **the system works out the new category by itself, and asks a human to approve it.**

A worked example

Five pneumatic cylinders were fed in — a type of product the system had never encountered. It had no idea what a pneumatic cylinder is. Left alone, it produced this description of the category, entirely from the five examples:

Property it discovered	What it concluded
Bore size	A measurement in millimetres, normally 0–101
Stroke length	A measurement in millimetres, normally 0–320
Operating pressure	A measurement in bar, normally 4–15
Cushioning	One of three options: adjustable, air, or rubber
Mounting	One of two options: basic or flange
Rod thread	One of two thread sizes

It also named the category “Pneumatic Cylinder”, and worked out which properties are essential rather than optional. It did this without being told anything, and without using AI — purely by noticing patterns across the five products.

Then it was tested properly:

- **Before learning**, a pneumatic cylinder was blocked from publication — the system correctly said it did not understand it.
- **After a human approved the new category**, the same product scored 99.7 out of 100.
- **A sixth cylinder**, never shown to it during learning, was classified correctly too. It had learned the *kind* of thing, not just those five items.
- **A deliberately absurd product** — a cylinder with a 50-metre bore — was rejected, caught by a rule the system had invented itself.

> **Why a human still has to press approve**
>
> The system had seen four products whose material was “bearing steel”, and proposed a rule that this was the *only* permitted material. Left unchecked, the first stainless steel rail would have been rejected for no good reason. A reviewer looking at the proposed rule spots that instantly. This is exactly why proposals are shown to a person in readable form rather than applied automatically.

6. Where artificial intelligence fits in

The system has two engines, and it chooses between them deliberately.

	The rules engine	The AI engine
Used for	Anything with a provable answer — standards lookups, unit conversion, arithmetic, contradiction checks	Anything requiring judgement — unfamiliar products, messy written descriptions, writing sales copy
Cost	Free	Costs money per product
Consistency	Identical answer every time	Can vary slightly
Can it be wrong?	Only if the underlying standard is wrong	Yes — so its answers are capped at a lower confidence

This split is a deliberate engineering choice, not a compromise. If an international standard fixes a bearing's bore at 25 mm, it would be worse to ask an AI to estimate it — the standard is simply correct, and it costs nothing to look up.

But the rules engine has a hard limit, and the project measured exactly where it is:

Situation	How much the rules engine recovers on its own
Product types backed by published standards	83%
Product types with no such standard	33%

That gap is precisely the AI's job. Where a rule exists, use the rule. Where no rule can exist, use judgement — and label it as judgement.

Seeing the AI without needing an account

There is a practical problem with demonstrating this. Using the AI engine normally requires the visitor to have their own paid account, which no reviewer will have.

So the AI was run once, in advance, over every demo product, and the results were saved into the application itself. Anyone visiting the live site can switch to the AI engine and see genuine AI-produced output — including its reasoning — without an account and at no cost to them. For example, on the bearing it explains:

> **The AI's own explanation of one value**
>
> “Part number 6205-2RS — under the ISO 15 bearing numbering standard, the last two digits ‘05’ multiplied by 5 give a 25 mm bore.”

That is the system showing its working, in a form a buyer could check for themselves.

7. Proof that it works

Claims are easy. This project was tested against 102 products where the correct answers were known in advance, so every claim below is measured rather than asserted.

How the test was built — and why it is fair

- For bearings and fasteners, the correct answers come from published international standards. Nobody involved in this project chose those numbers, so the system cannot have been quietly tuned to match them.
- Each product was tested twice: once with most of its information removed, to see how much the system could recover; and once with a deliberate error planted in it, to see whether the error was caught.
- Only the information that was *hidden* counts towards the score. Credit is not taken for repeating back what it was already told.

What was measured	Result	What it means
Information recovered	2.75×	Nearly three times as much usable information came out as went in
Hidden details found	61%	Of the details deliberately removed, it worked out six in ten
Wrong answers stated confidently	**0%**	**Of every value it asserted with evidence, not one contradicted the truth**
Planted errors caught	100%	All 51 deliberate mistakes were detected
False alarms	0%	It never cried wolf on a clean product
Speed	340 per second	A 200,000-product catalogue in about ten minutes

> **Which number actually matters**
>
> Not the 2.75×. Any system can fill every box by inventing content — that scores brilliantly on “how much did you produce” and is worthless. The number that matters is the **0%**: of everything the system stated as fact, none of it was wrong. The gaps it left were honest gaps.

Being honest about the weaker points

- The 0% applies to values backed by evidence. Values the system marks as “typical, please confirm” are right about three times in four — which is why they are visibly marked rather than presented as fact.
- For product types with no published standard, the test data was written by hand rather than taken from an external source. Those results are reported separately and not blended into the headline.
- Scanned documents — photographs of paper — cannot be read. The system detects this and says so rather than returning nothing.

8. What exists right now

Item	Status
Working prototype	**Live on the internet**, monitored every five minutes, 100% uptime
Source code	Complete and version-controlled, with the reasoning behind each decision recorded
Automated tests	Five test suites, all passing, none of which cost anything to run
Measured results	Published, with the method open to inspection
Presentation deck	Not started — awaiting the required template
Demo video	Not started

What was built, and when

All of the following was built in a single working day, roughly four days ahead of the original plan.

Stage	What it delivered
Foundation	The eight-stage production line, ten product categories, sixteen engineering checks, and the website
Measurement	The 102-product test. Building it exposed three genuine faults in the system, all fixed.
Documents	Reading datasheets and supplier web pages, handling three different PDF layouts
Learning	Working out new product categories automatically, with human approval
Deployment	Packaged as a single unit, published to the internet, monitored

A note on cost control

Using AI costs real money per product, and early in the project a mistake caused about $2.60 to be spent with nothing to show for it: two identical jobs were accidentally started at once and both were lost before finishing.

The response was to make that failure impossible rather than to be more careful. Three safeguards were built and tested:

- A second job now **refuses to start** while one is already running.
- Spending is checked **after every single product**, and the job stops the instant it crosses a limit you set.
- Completed work is saved, so a stopped job **resumes for free** rather than starting again.

Total spent on the project so far: **about $0.60**. The published website is deliberately incapable of spending anything at all, no matter how many people use it.

9. What is left

Task	Notes
Presentation deck	Blocked until the required template is downloaded from the submission portal
Demo video	A short walkthrough of the live site
Finish the AI comparison	A partially-completed measurement of exactly how much the AI adds over the rules engine. Roughly $2 to finish.
Make the code repository public	Required at submission; it is private while under development

Honest assessment

The engineering is in good shape and comfortably ahead of schedule. The remaining risk is not technical — it is that the work is currently better than the explanation of it. A reviewer will spend a few minutes forming a judgement, and right now there is no deck or video to shape that judgement.

The sensible course from here is to stop adding capability and spend the remaining time making what exists easy to understand quickly.

Glossary

Terms used in this document, in plain English.

Term	Meaning
Attribute	One fact about a product — its width, its material, its voltage.
Bore	The diameter of the hole through the middle of a part such as a bearing.
Category / taxonomy	The filing system that decides what kind of thing a product is. It determines which facts are worth collecting.
Confidence	How sure the system is about a value, shown as a percentage. Low confidence is a request for a human to check.
Enrichment	Filling in missing product information.
ISO 15 / ISO 898-1 / SAE J517	Published international engineering standards. They define, for example, the exact dimensions a bearing labelled 6205 must have.
Provenance	The record of where a piece of information came from — the receipt attached to every value.
Readiness score	A mark out of 100 combining how complete a record is, how confident, and how many problems were found.
Validation	Checking data for errors, contradictions and impossibilities.
Verdict	The final decision on a product: publish (ready to sell), review (needs a human), or blocked (must not be published).

Live prototype: industrial-product-intelligence.onrender.com

Document generated from the project's own records.